

ELEC 442

Lab 0

GOALS

- ☐ *Install Core Software*
- ☐ *Install Quanser Software*
- ☐ *Test Software*

1 Environment Setup

The following guides in the Lab 0 repo give detailed instructions. **If you need help please ask an instructor/TA. For Lab 0, no submission is required.**

1. [Software Installation Guide \(personal_software_installation.md\)](#)

- https://github.com/UBC-ELEC-442-Labs-2026/Lab-0/blob/main/setup_instructions/personal-software-installation.md

2. [Computer Setup Guide \(personal_computer_setup.md\)](#)

- https://github.com/UBC-ELEC-442-Labs-2026/Lab-0/blob/main/setup_instructions/personal-computer-setup.md

1.1 Setup Checklist

Before walking into the lab, ensure you have:

- ☐ Read through and followed the guides
 - ☐ *Software Installation Guide*
 - ☐ *Computer Setup Guide*
- ☐ Installed core software
 - ☐ *Git*
 - ☐ *Python*
 - ☐ *VS Code*
- ☐ Prepared Virtual Arm (Windows Only)
 - ☐ *Installed QLABs*
 - ☐ *Created Quanser Account (and provided it to the instructor)*

2 Environment Test

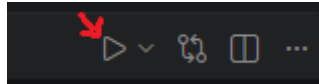
2.1 Core Software Test

2.1.1 Git and VS Code

Try cloning the Lab 0 GitHub Repo to your device. Then try opening it in VS Code. Follow the the Computer Setup Guide and ask instructors for help if issues arise.

2.1.2 Python

With Lab 0 open in VS Code, open `Lab_0-python_test.py` and try to run it. This can be done by pressing the “run” button in the top right of VS Code (see below)



A terminal pane should pop out and display the following output (numpy and python versions could cause the number to differ slightly, but it should be ≈ 0):

```
hello world
1.2246467991473532e-16
```

Alternatively, you can open a terminal and enter:

```
python replace_this_with_path_to_python_file
```

2.2 Q Labs/Virtual Arm Test

1. Open “Quanser Interactive Labs”
2. Sign in (or create a new account and let the instructors know)
3. Select “QArm” > “QArm Workspace”
4. Run `Lab_0-QLabs_test.py` from the Lab 0 Repo

If successful, you should see the virtual QArm move between an outstretched position and a neutral position every three seconds for 10 cycles.

Congrats! You now have all required software setup, and you’re ready for the rest of the labs!

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